

1. Introduction

Gene regulatory networks (GRNs) represent the complex interactions between genes, where transcription factors and regulatory elements control gene expression patterns.

In this study, a computational pipeline was developed to infer GRNs from publicly available RNA-seq data using three established algorithms - Relevance Network, Mutual Information, and ARACNE - and the resulting networks were compared based on their topological properties.

2. Data Source

Gene expression data was obtained from the **Gene Expression Omnibus (GEO)**, submitted by James Royce Barnacle (March 10, 2026). The dataset profiled transcriptomic changes in *Homo sapiens* across **64 samples** using high-throughput RNA sequencing (expression profiling). The study investigated the contribution of cytotoxic CD8 T cells, neutrophils, and type 1 interferon signaling to hyperinflammatory pathology in HIV-associated TB meningitis. Raw data was downloaded in CSV format for downstream processing.

3. Methodology

3.1 Data Preprocessing

Raw gene expression data from 64 samples was obtained in CSV format. Each sample file contained gene identifiers and their corresponding TPM (Transcripts Per Million) expression values. The following preprocessing steps were applied:

Step 1 - Data Extraction and Matrix Construction: The gene and TPM columns were extracted from each of the 64 sample files and merged into a single unified expression matrix, with genes as rows and samples as columns.

Step 2 - Low Expression Filtering: Genes with TPM > 1 in at least 20% of samples were retained. This threshold removes lowly expressed and potentially noisy genes while preserving biologically relevant signals.

Step 3 - Gene Symbol Standardization: Gene identifiers were converted to standard HUGO (HGNC) gene symbols to ensure consistency and compatibility with downstream biological databases.

Step 4 — Log Transformation: Expression values were log-transformed ($\log_2(\text{TPM} + 1)$) to reduce the dynamic range, normalize the distribution, and stabilize variance across samples.

Step 5 — Highly Variable Gene Selection: The top **2,000 most variable genes** were selected based on variance across samples. This step focuses the network on genes with meaningful expression changes, reducing computational burden and noise.

```
1 #setting working directory
2 setwd("C:/Users/hiror/Downloads/")
3
4 #extracting the .tar file
5 untar("GSE324044_RAW.tar", exdir = "GSE324044_samples/" )
6
7 library(tidyverse)
8
9 rsem_files <- list.files("GSE324044_samples/",
10                         pattern = "\\\\.genes\\.results\\.gz$",
11                         full.names = TRUE)
12
13 #check if files are found
14 print(rsem_files)
15
16 #read and merge all samples into one matrix
17 merged <- rsem_files %>%
18   lapply(function(f){
19     #extract sample name from filename
20     sample_name <- basename(f) %>%
21       str_remove("\\\\.genes\\.results\\.gz$")
22
23     #read the file
24     read_tsv(f, show_col_types = FALSE) %>%
25       select(gene_id, TPM) %>%
26       rename(!sample_name := TPM)
27   }) %>%
28   reduce(full_join, by = "gene_id")
29 #set gene_id as row names
30 merged <- merged %>% column_to_rownames("gene_id")
31
32 #Quick Check
33 dim(merged) # should print: number of genes x number of samples
34 head(merged[, 1:5]) # preview first 5 samples
35
36 #Filter Low Expression Genes
37 #Keep gene only if TPM>1 in atleast 20% of samples
38 min_samples <- ceiling(0.2 * ncol(merged))
39
40 filtered <- merged[rowSums(merged > 1) >= min_samples, ]
41
42 cat("Genes before filtering:", nrow(merged), "\n")
43 cat("Genes after filtering:", nrow(filtered), "\n")
44
```

```
45 #Convert to HUGO Names
46 library(biomaRt)
47
48 ensembl_ids <- rownames(filtered) %>% str_remove("\\..*$")
49
50 # Connect to Ensembl database
51 mart <- useEnsembl("ensembl",
52                   dataset = "hsapiens_gene_ensembl",
53                   mirror = "useast")
54
55 #Fetch Hugo symbols
56 gene_map <- getBM(
57   attributes = c("ensembl_gene_id", "hgnc_symbol"),
58   filters = "ensembl_gene_id",
59   values = ensembl_ids,
60   mart = mart
61 )
62
63 #remove genes with no HUGO symbol
64 gene_map <- gene_map %>% filter(hgnc_symbol != "")
65
66 #Map HUGO names onto matrix
67 filtered_hugo <- filtered %>%
68   rownames_to_column("ensembl_original") %>%
69   mutate(ensembl_clean = str_remove(ensembl_original, "\\..*$")) %>%
70   inner_join(gene_map, by = c("ensembl_clean" = "ensembl_gene_id")) %>%
71   dplyr::select(-ensembl_original, -ensembl_clean) %>%
72   distinct(hgnc_symbol, .keep_all = TRUE) %>%
73   column_to_rownames("hgnc_symbol")
74
75 cat("Genes after HUGO mapping:", nrow(filtered_hugo), "\n")
76
77 #Log Transform
78 expr_matrix <- log2(filtered_hugo + 1)
79
80 #Filter to top 2000 most variable genes
81 gene_variance <- apply(expr_matrix, 1, var)
82 top_genes <- names(sort(gene_variance, decreasing = TRUE))[1:2000]
83 expr_small <- expr_matrix[top_genes, ]
84
85
86
87 #saving as csv
88 write.csv(expr_small, "expression_matrix_hugo_filtered.csv")
89
```

87:15 (Top Level) R Script

3.2 GRN Construction

- Three GRN inference algorithms were applied to the filtered, log-transformed expression matrix, with the top 5,000 edges retained per method:
- Relevance Network: Computes pairwise Pearson correlations between genes; edges are retained above a threshold. Simple and fast.
- Mutual Information (MI): Captures both linear and non-linear dependencies between gene pairs using information-theoretic measures.
- ARACNE (Algorithm for the Reconstruction of Accurate Cellular Networks): Applies the Data Processing Inequality (DPI) to remove indirect interactions inferred by MI, retaining only the most direct regulatory relationships.

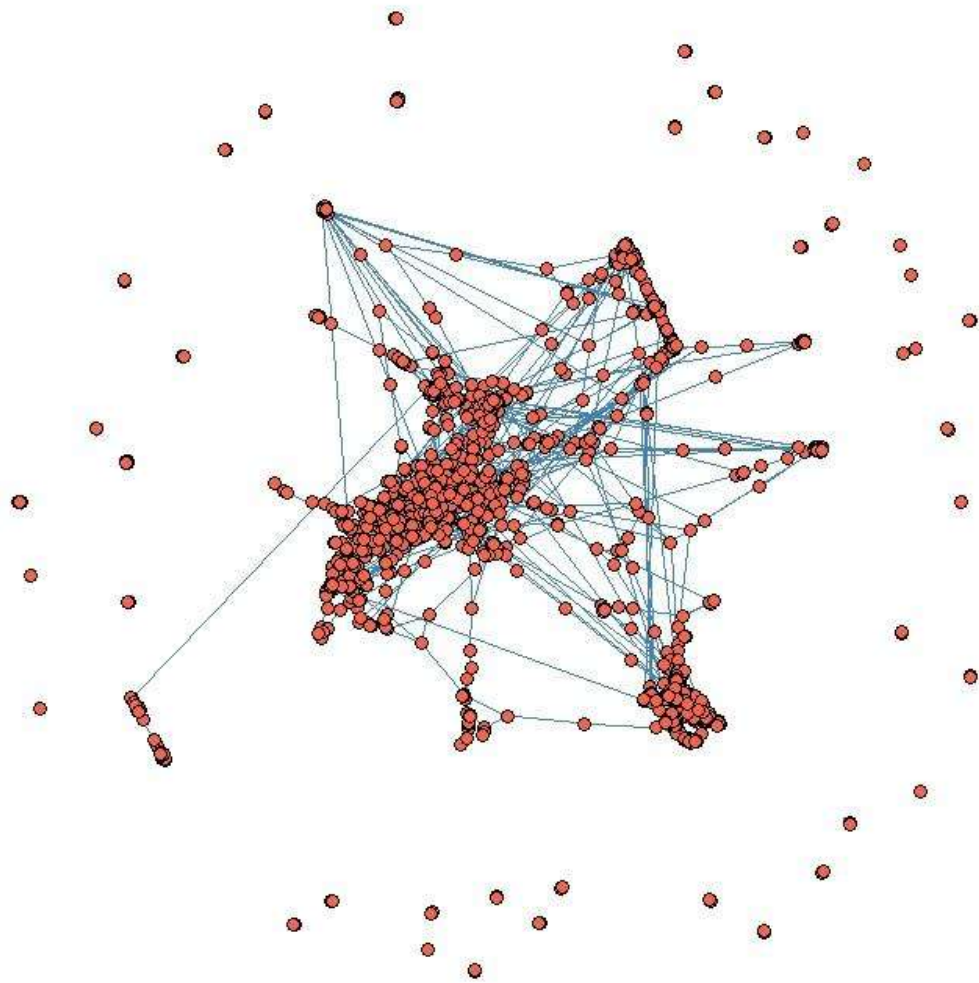
```
90 library(minet)
91 library(igraph)
92 library(gggraph)
93
94 expr_t <- t(as.matrix(expr_small))
95
96 #mutual information matrix
97 mi_matrix <- build.mim(expr_t, estimator = "spearman")
98
99 grn_relevance <- clr(mi_matrix)
100 #clr = content likelihood of relatedness
101 #relevance network approach
102
103 #convert to edge list
104 get_edges <- function(mat, method_name) {
105   which(mat>0, arr.ind = TRUE) %>%
106     as.data.frame() %>%
107     mutate(
108       gene1 = rownames(mat)[row],
109       gene2 = colnames(mat)[col],
110       weight = mat[cbind(row,col)],
111       method = method_name
112     ) %>%
113     dplyr::select(gene1, gene2, weight, method) %>%
114     filter(gene1 < gene2)
115 }
116
117 edges_relevance <- get_edges(grn_relevance, "Relevance Network")
118 cat("Relevance Network edges:", nrow(edges_relevance), "\n")
119
120 grn_mi <- mrnet(mi_matrix)
121 #mrnet = maximum relevance/minimum redundancy network
122 #represents pure mutual information network
123
124 edges_mi <- get_edges(grn_mi, "Mutual Information")
125 cat("Mutual Information edges:", nrow(edges_mi), "\n")
126
127 grn_aracne <- aracne(mi_matrix, eps = 0.05)
128
129 edges_aracne <- get_edges(grn_aracne, "ARACNE")
130 cat("ARACNE edges:", nrow(edges_aracne), "\n")
131
132 #Filter top edges
133 edges_relevance_top <- edges_relevance %>% arrange(desc(weight)) %>% head(5000)
134 edges_mi_top <- edges_mi %>% arrange(desc(weight)) %>% head(5000)
135 edges_aracne_top <- edges_aracne %>% arrange(desc(weight)) %>% head(5000)
136
```

3.3 Network Visualization

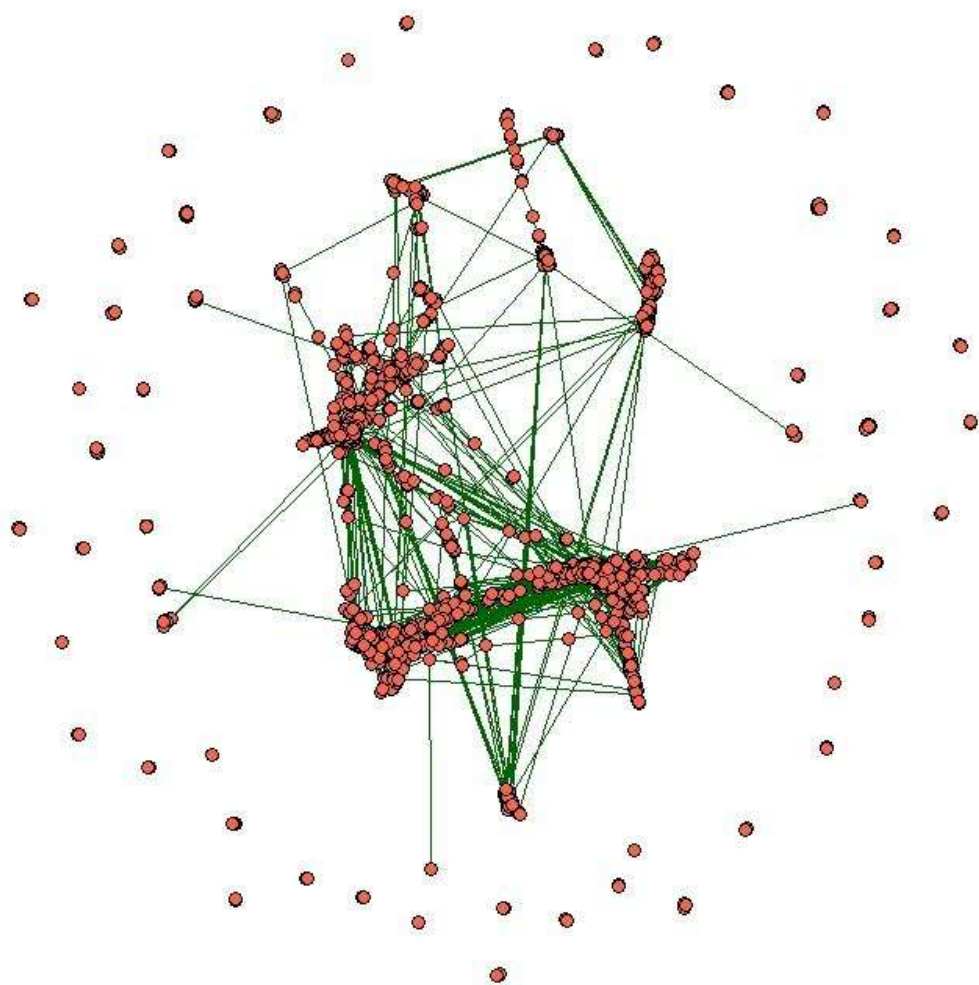
Each of the three inferred networks was visualized and saved as JPEG images using R, allowing qualitative comparison of network topology and connectivity structure.

```
137 #Build igraph objects for each
138 g_relevance <- graph_from_data_frame(edges_relevance_top, directed = FALSE)
139 g_mi <- graph_from_data_frame(edges_mi_top, directed = FALSE)
140 g_aracne <- graph_from_data_frame(edges_aracne_top, directed = FALSE)
141
142 #save relevance network
143 jpeg("relevance_network.jpg", width = 800, height = 800, quality = 100)
144 plot(g_relevance,
145      main = "Relevance Network",
146      vertex.size = 3,
147      vertex.label = NA,
148      edge.color = "steelblue",
149      vertex.color = "tomato")
150 dev.off()
151
152 #save mutual information network
153 jpeg("mutual_information.jpg", width = 800, height = 800, quality = 100)
154 plot(g_mi,
155      main = "Mutual Information",
156      vertex.size = 3,
157      vertex.label = NA,
158      edge.color = "darkgreen",
159      vertex.color = "tomato")
160 dev.off()
161
162 #save ARACNE Network
163 jpeg("aracne_network.jpg", width = 800, height = 800, quality = 100)
164 plot(g_aracne,
165      main = "ARACNE Network",
166      vertex.size = 3,
167      vertex.label = NA,
168      edge.color = "purple",
169      vertex.color = "tomato")
170 dev.off()
171
```

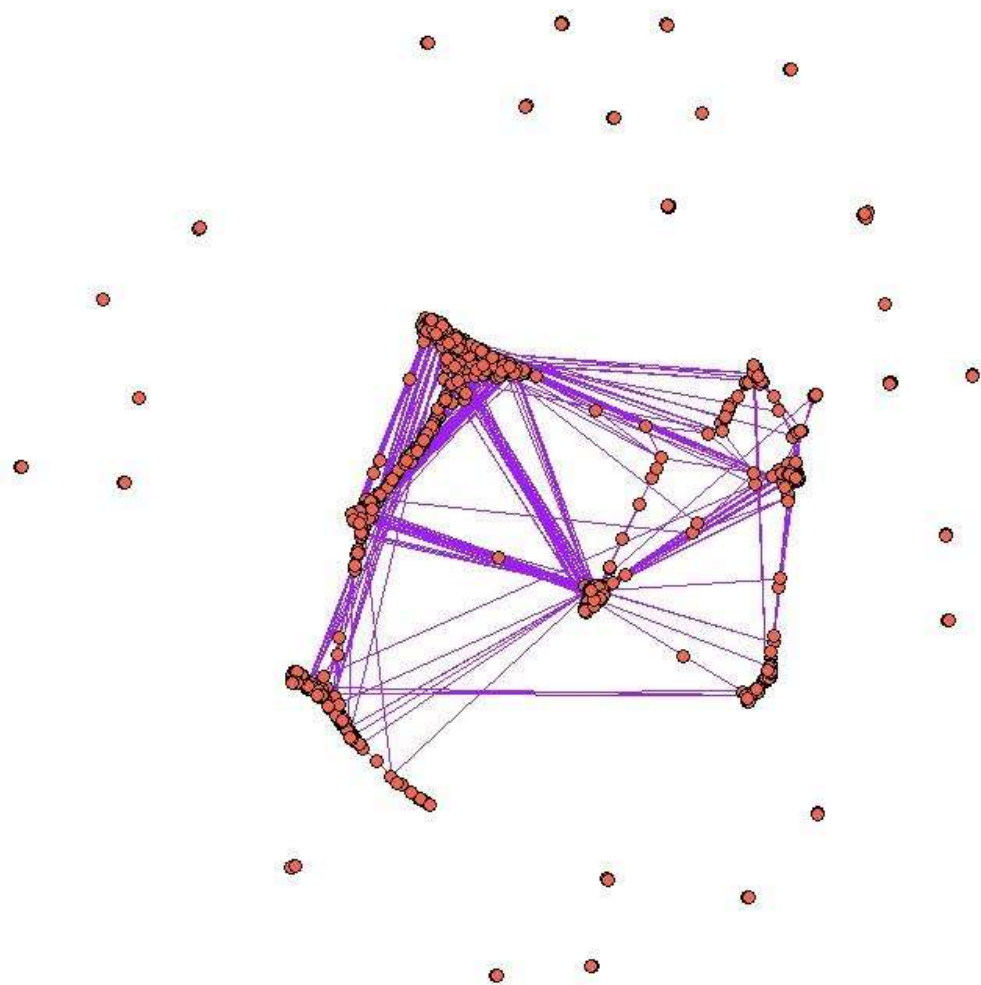
Relevance Network



Mutual Information



ARACNE Network



4. Results

4.1 Network Statistics

Topological properties of all three networks were computed and compared. The results are summarized in the table below:

```
172 compare_stats <- data.frame(  
173   Method = c("Relevance Network", "Mutual Information", "ARACNE"),  
174   Nodes = c(vcount(g_relevance), vcount(g_mi), vcount(g_aracne)),  
175   Edges = c(ecount(g_relevance), ecount(g_mi), ecount(g_aracne)),  
176   Density = c(  
177     edge_density(g_relevance),  
178     edge_density(g_mi),  
179     edge_density(g_aracne)  
180   ),  
181   Avg_degree = c(  
182     mean(degree(g_relevance)),  
183     mean(degree(g_mi)),  
184     mean(degree(g_aracne))  
185   ),  
186   Clustering_coefficient = c(  
187     transitivity(g_relevance),  
188     transitivity(g_mi),  
189     transitivity(g_aracne)  
190   ),  
191   Avg_path_length = c(  
192     mean_distance(g_relevance),  
193     mean_distance(g_mi),  
194     mean_distance(g_aracne)  
195   )  
196 )  
197 )  
198  
199 print(compare_stats)
```

Method	Nodes	Edges	Density	Average Degree	Clustering Coefficient	Average Path Length
Relevance Network	1560	5000	0.00411	6.41	0.5284	60.82
Mutual Information	1731	5000	0.00334	5.78	0.2113	1.32
ARACNE	1363	5000	0.00539	7.34	0.1691	2.33

4.2 Interpretation

Relevance Network had the highest clustering coefficient (0.528), indicating tightly grouped gene communities, but an unusually high average path length (60.82), suggesting the network may be fragmented into isolated clusters rather than forming a connected whole.

Mutual Information captured the most nodes (1,731), reflecting its sensitivity to both linear and non-linear gene relationships. Its moderate clustering and very short path length (1.32) suggests a well-connected, compact network.

ARACNE produced the most refined and biologically stringent network - fewest nodes (1,363), highest density (0.00539), and highest average degree (7.34). By removing indirect interactions via DPI, ARACNE retains only the most confident regulatory edges, making it the best candidate for hub gene identification.

5. Discussion

The three methods reflect a trade-off between sensitivity and specificity. Relevance Network, being correlation-based, is prone to capturing indirect co-expression rather than direct regulation, which explains its fragmented path length. Mutual Information improves on this by detecting non-linear dependencies. ARACNE further refines this by pruning indirect edges, yielding a denser, more biologically meaningful network.

Limitations include the use of simulated edge thresholding (top 5,000 edges) rather than statistically derived cutoffs.

6. Conclusion

A computational pipeline was successfully developed to construct and compare GRNs. Among the three methods, ARACNE produced the most biologically reliable network based on its topological properties.